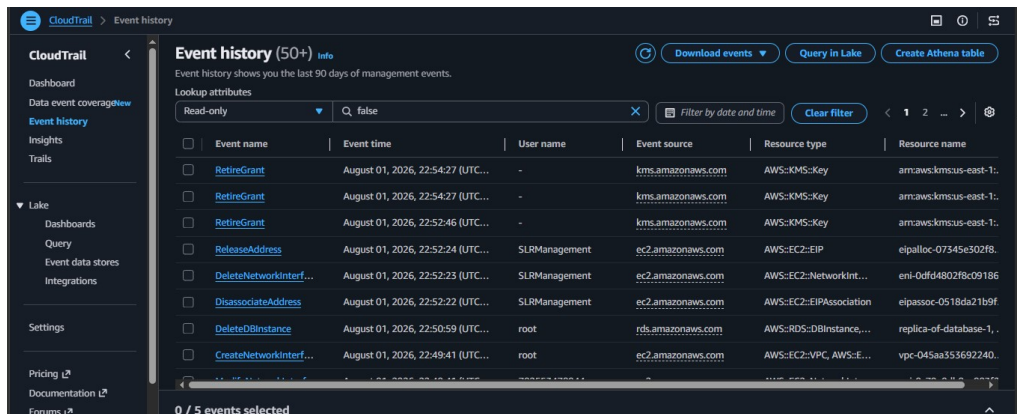


Cloud Trails with Athena analyzing tool

1.What is Cloud Trails ? configure a S3 bucket for cloudtrails.

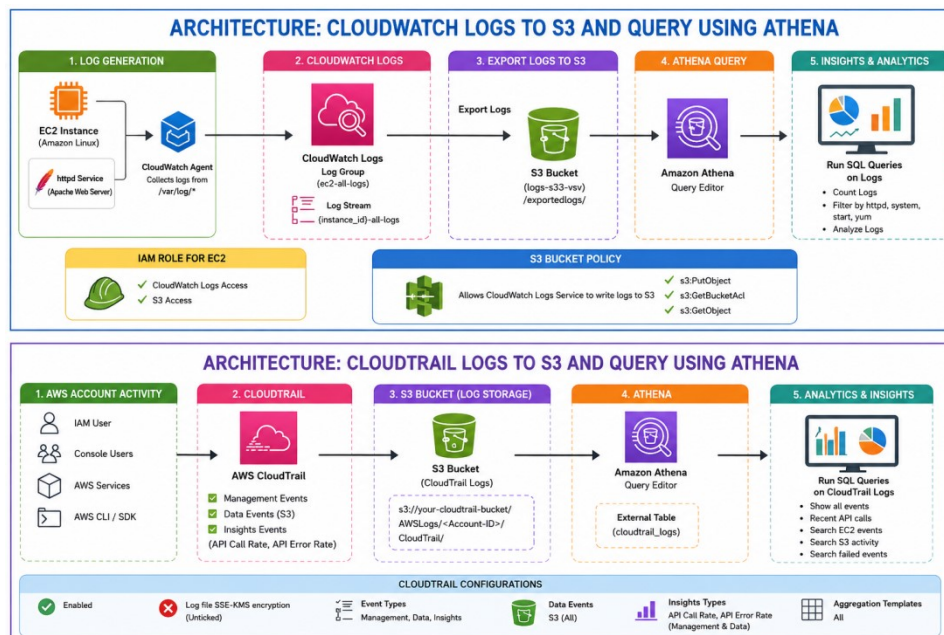
1. Cloud trail is an aws service used to store activity logs and api call history .
2. The retention period of logs is 90 days
3. We can configure cloud trails to store the logs permanently using S3 bucket.
4. To efficiently use/handle millions and billions of logs , we can integrate Athena analytics service to analysis the logs .
5. Reference - [itmukesh-16/Athena](#)



The screenshot shows the AWS CloudTrail 'Event history' console. It displays a table of events with columns: Event name, Event time, User name, Event source, Resource type, and Resource name. The events listed include RetireGrant, ReleaseAddress, DeleteNetworkInterface, DisassociateAddress, DeleteDBInstance, and CreateNetworkInterface. The console also features a sidebar with navigation options like Dashboard, Data event coverage, Event history, Insights, Trails, Lake, Dashboards, Query, Event data stores, Integrations, Settings, Pricing, Documentation, and Forums.

Event name	Event time	User name	Event source	Resource type	Resource name
RetireGrant	August 01, 2026, 22:54:27 (UTC...)	-	kms.amazonaws.com	AWS::KMS-Key	arn:aws:kms-east-1:...
RetireGrant	August 01, 2026, 22:54:27 (UTC...)	-	kms.amazonaws.com	AWS::KMS-Key	arn:aws:kms-east-1:...
ReleaseAddress	August 01, 2026, 22:52:46 (UTC...)	-	kms.amazonaws.com	AWS::KMS-Key	arn:aws:kms-east-1:...
ReleaseAddress	August 01, 2026, 22:52:24 (UTC...)	SLRManagement	ec2.amazonaws.com	AWS::EC2-EIP	eipalloc-07345e302f8...
DeleteNetworkInterf...	August 01, 2026, 22:52:23 (UTC...)	SLRManagement	ec2.amazonaws.com	AWS::EC2-NetworkInt...	eni-0df4802f8c09186...
DisassociateAddress	August 01, 2026, 22:52:22 (UTC...)	SLRManagement	ec2.amazonaws.com	AWS::EC2-EIPAssociation	eipassoc-0518da21b9f...
DeleteDBInstance	August 01, 2026, 22:50:59 (UTC...)	root	rds.amazonaws.com	AWS::RDS-DBInstance...	replica-of-database-1...
CreateNetworkInterf...	August 01, 2026, 22:49:41 (UTC...)	root	ec2.amazonaws.com	AWS::EC2-VPC, AWS::E...	vpc-045aa53692240...

Store CloudTrail data into S3 bucket also integrate Athena to analyze the logs .



Open Cloud Trails > trails > Create trails.

The screenshot shows the 'Create trail' page in the AWS CloudTrail console. The trail name is 'Daily-activity-logs'. The storage location is set to 'Create new S3 bucket'. The trail log bucket and folder is 'aws-cloudtrail-logs-198302589656-d1190492'. Log file S3-KMS encryption is enabled. Additional settings include 'Log file validation' and 'SNS notification delivery', both of which are enabled.

Trail name
Enter a display name for your trail.
Daily-activity-logs
3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

Storage location
Create new S3 bucket
Create a bucket to store logs for the trail.

Trail log bucket and folder
Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.
aws-cloudtrail-logs-198302589656-d1190492
Logs will be stored in aws-cloudtrail-logs-198302589656-d1190492/AWSLogs/198302589656

Log file S3-KMS encryption
Enabled

Additional settings
Log file validation
Enabled
SNS notification delivery
Enabled

Select all required options

Now go to S3 bucket to check if the log bucket created or not

The screenshot shows the 'Trails' page in the AWS CloudTrail console. The table lists the trail 'Daily-activity-logs' with details such as Home region (US East (N. Virginia)), Multi-region trail (Yes), ARN (arn:aws:cloudtrail:us-east-1:198302589656:trail/Daily-activity-logs), Insights (Enabled), Organization trail (No), S3 bucket (aws-cloudtrail-logs-198302589656-d1190492), Log file prefix (-), and CloudWatch Logs log group (-).

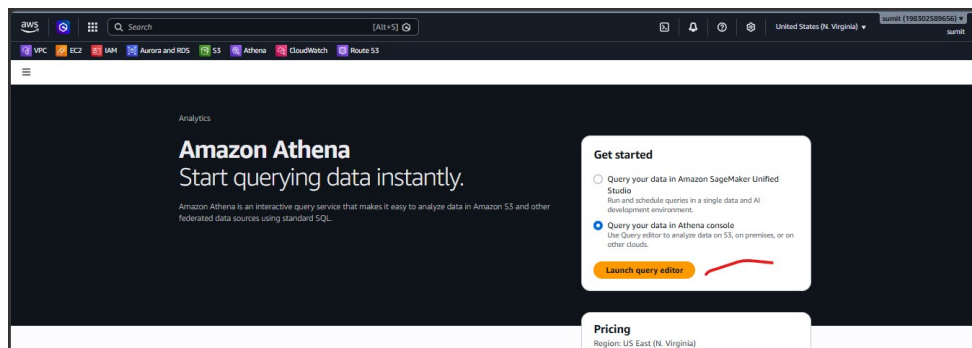
Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group
Daily-activity-logs	US East (N. Virginia)	Yes	arn:aws:cloudtrail:us-east-1:198302589656:trail/Daily-activity-logs	Enabled	No	aws-cloudtrail-logs-198302589656-d1190492	-	-

We can see the logs stored into S3 bucket.

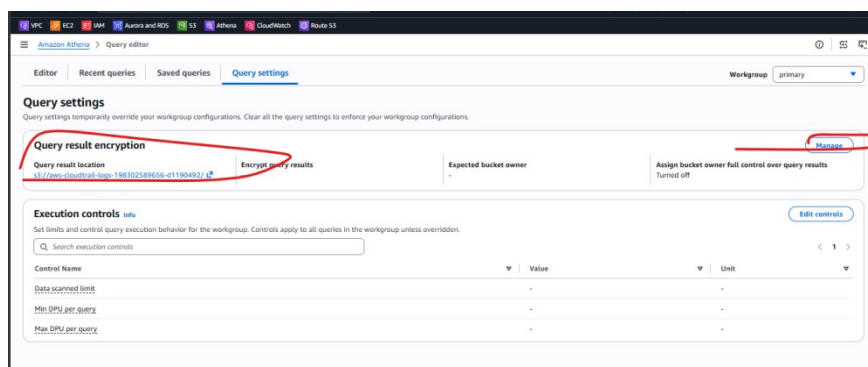
The screenshot shows the Amazon S3 console for the bucket 'aws-cloudtrail-logs-198302589656-dd97cad7'. The 'Objects' tab is selected, showing a list of folders: 'CloudTrail-Digest/', 'CloudTrail-Insight/', and 'CloudTrail/'.

Name	Type	Last modified	Size	Storage class
CloudTrail-Digest/	Folder	-	-	-
CloudTrail-Insight/	Folder	-	-	-
CloudTrail/	Folder	-	-	-

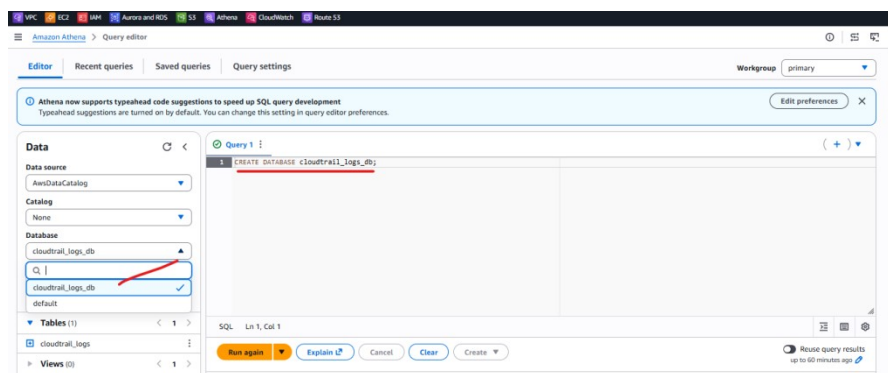
Now integrate Athena for analyzing logs (Athena doesn't store data it only fetch data from source and analyze the data ,it only charges per query cost . it has max duration of 15min)



Choose query settings tab > manage > give bucket name



Go to Editor > create a database (it doesn't store any actual data, its just reference for logs)



Create a schema , its basically log structure ,ie tells how data stored in s3 bucket (column names , rows,foreign keys etc).open a new tab and run the query . make sure you are giving s3 bucket name and account id

Athena now supports typeahead code suggestions to speed up SQL query development. Typeahead suggestions are turned on by default. You can change this setting in query editor preferences. [Edit preferences](#)

Data

Data source: **AwsDataCatalog**

Catalog: **None**

Database: **cloudtrail_logs_db**

Tables and views [Create](#)

Filter tables and views

Tables (1)

- cloudtrail_logs

Views (0)

Query 1 **Query 2** **Query 3** **Query 4**

```

34 requestParameters:STRING,
35 responseElements:STRING,
36 additionalEventData:STRING,
37 requestId:STRING,
38 eventId:STRING,
39 readonly:STRING,
40 eventType:STRING,
41 managementEvent:STRING,
42 recipientAccountId:STRING,
43 eventCategory:STRING
44 >
45 >
46 )
47 ROW FORMAT SERDE 'org.openx.data.jsonserde.JsonSerDe'
48 LOCATION 's3://aws-logs-198302589656-d1190492/AmazonLogs/198302589656/CloudTrail/';

```

SQL Ln 48, Col 78

[Run again](#) [Explain](#) [Cancel](#) [Clear](#) [Create](#)

Query results **Query stats**

Completed Time in queue: 84 ms Run time: 278 ms Data scanned: -

Now open new query tab and run a sql query to fetch the log data of a particular date

Data

Data source: **AwsDataCatalog**

Catalog: **None**

Database: **cloudtrail_logs_db**

Tables and views [Create](#)

Filter tables and views

Tables (1)

- cloudtrail_logs

Views (0)

Query 1 **Query 2** **Query 3** **Query 4**

```

1 SELECT
2 record.eventTime,
3 record.eventSource,
4 record.eventName,
5 record.eventSourceAddress,
6 FROM cloudtrail_logs
7 CROSS JOIN UNNEST(Records) AS t(record)
8 WHERE record.eventTime > DATE '2025-08-15'
9 ORDER BY record.eventTime DESC;

```

SQL Ln 9, Col 25

[Run again](#) [Explain](#) [Cancel](#) [Clear](#) [Create](#)

Query results **Query stats**

Completed Time in queue: 97 ms Run time: 1.133 sec Data scanned: 757.90 KB

Results (5,160) [Copy](#) [Download results CSV](#)

#	eventTime	eventSource	eventName	sourceIPAddress
1	2025-08-15T12:29:48Z	notifications.amazonaws.com	ListManagedNotificationEvents	49.238.55.245
2	2025-08-15T12:29:00Z	glue.amazonaws.com	GetCatalogs	49.238.55.245
3	2025-08-15T12:28:59Z	ic.amazonaws.com	GetObject	athena.amazonaws.com

Task Over